Benchmarking Superconducting Qubits with Randomized Benchmarking in Qiskit

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***Abstract*—This project investigates the fidelity of quantum gates in superconducting qubits using Randomised Benchmarking (RB) and Interleaved Randomised Benchmarking (IRB). Experiments will be implemented in Qiskit and executed both on noisy simulators with device-calibrated noise models and on real IBM Quantum hardware. The study aims to extract the error per Clifford (EPC) for single- and two-qubit gates, analyse discrepancies between simulation and hardware, and relate these errors to physical noise mechanisms such as decoherence times (T_1, T_2) and readout errors. Results will provide quantitative insight into hardware limitations and contribute to the understanding of gate performance in near-term quantum devices, while demonstrating practical implementation of quantum benchmarking protocols.**

I. Introduction

SUPERCONDUCTING qubits are among the leading platforms for near-term quantum computing, offering fast gate operations and scalability potential. However, their performance is limited by decoherence, control errors, and readout imperfections. Quantifying the fidelity of quantum gates is therefore essential to understand hardware limitations and to guide optimisation strategies.

Randomised Benchmarking (RB) is a widely used protocol to measure the average error per Clifford gate in a scalable and robust manner, mitigating the effects of state preparation and measurement errors. Interleaved RB (IRB) extends this approach to estimate the fidelity of specific gates, such as CNOT, which are critical for universal quantum computation.

In this project, we implement RB and IRB using Qiskit and Qiskit Experiments, executing experiments both in simulators with noise models extracted from IBM Quantum devices and on real superconducting hardware. This approach allows us to compare idealised noise models with actual hardware performance, relate errors to physical mechanisms such as T_1, T_2, and readout noise, and provide insights into the reliability of quantum operations in near-term devices.

There are various hardware platforms for qubit implementation. As of 2025, the leading technologies for quantum processors, in terms of scalability, use neutral atoms in optical lattices (Atom Computing, >1200 qubits [1]), superconducting qubits (IBM, *Condor*, 1121 qubits [2]), and photonics measured-based (Xanadu, *Borealis*, 216 qubits [3]). In terms of fidelity, however, trapped ion technologies seem to be leading the industry (Quantinuum, *H2*, 99.997% - 1 qubit, 99.87% - 2 qubit [4]).

Scalability vs Fidelity

While scalability is a classical concept, it gains new dimension when used in the context of quantum computing. Because qubits rely on nanoscale phenomena, they are highly susceptible to interference, both from environmental noise and neighbouring qubits. For this reason, quantifying the maximum number of qubits that can operate simultaneously without performance loss is a crucial factor in system design. [5].

Fidelity characterizes the agreement between the actual outcome of the operation and the desired state. It generally depends on the initial state on which the operation is applied. Since the operation may generally be applied to any arbitrary state, it is natural to average the fidelity over all pure initial states, chosen uniformly in the system's Hilbert space [6].

II. Simulation with noise Models

A. Interleaved Randomised Benchmarking

Randomized Benchmarking proves to be advantageous over different qubit characterisation techniques, such as Quantum Process Tomography. While RB is scalable in terms of number of qubits comprising the system, it is also to provide a noise measure independent of state preparation and measurement errors (SPME) [7].

B. 0000000

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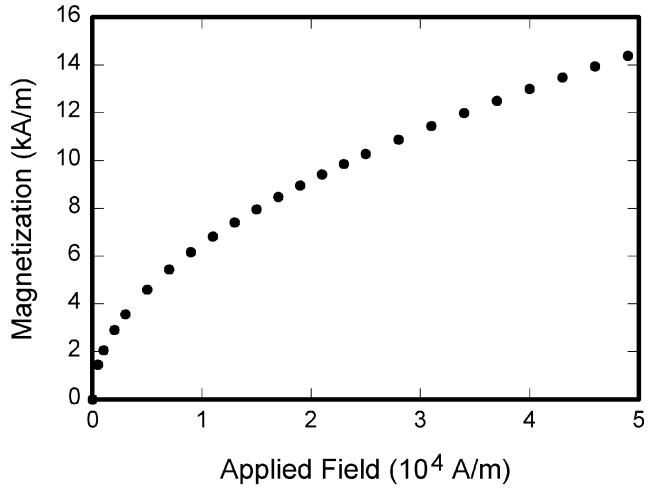

Fig. 1. Magnetization as a function of applied field. Note that "Fig." is abbreviated. There is a period after the figure number, followed by two spaces. It is good practice to explain the significance of the figure in the caption.

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TABLE I
UNITS FOR MAGNETIC PROPERTIES

Symbol	Quantity	Conversion from Gaussian and CGS EMU to SI [a]
Φ	magnetic flux	1 Mx → 10^{-8} Wb = 10^{-8} V·s
B	magnetic flux density, magnetic induction	1 G → 10^{-4} T = 10^{-4} Wb/m^2
H	magnetic field strength	1 Oe → $10^3/(4\pi)$ A/m
m	magnetic moment	1 erg/G = 1 emu → 10^{-3} A·m^2 = 10^{-3} J/T
M	magnetization	1 erg/(G·cm^3) = 1 emu/cm^3 → 10^3 A/m
$4\pi M$	magnetization	1 G → $10^3/(4\pi)$ A/m
σ	specific magnetization	1 erg/(G·g) = 1 emu/g → 1 A·m^2/kg
j	magnetic dipole moment	1 erg/G = 1 emu → $4\pi \times 10^{-10}$ Wb·m
J	magnetic polarization	1 erg/(G·cm^3) = 1 emu/cm^3 → $4\pi \times 10^{-4}$ T
χ, κ	susceptibility	1 → 4π
χ_ρ	mass susceptibility	1 cm^3/g → $4\pi \times 10^{-3}$ m^3/kg
μ	permeability	1 → $4\pi \times 10^{-7}$ H/m = $4\pi \times 10^{-7}$ Wb/(A·m)
μ_r	relative permeability	$\mu \rightarrow \mu_r$
w, W	energy density	1 erg/cm^3 → 10^{-1} J/m^3
N, D	demagnetizing factor	1 → $1/(4\pi)$

Vertical lines are optional in tables. Statements that serve as captions for the entire table do not need footnote letters.

[a]Gaussian units are the same as cgs emu for magnetostatics; Mx = maxwell, G = gauss, Oe = oersted; Wb = weber, V = volt, s = second, T = tesla, m = meter, A = ampere, J = joule, kg = kilogram, H = henry.

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$$\int_0^{r_2} F(r,\varphi)\, dr\, d\varphi = [\sigma r_2 / (2\mu_0)] \cdot \int_0^{\infty} \exp(-\lambda\, |z_j - z_i|)\, \lambda^{-1} J_1(\lambda r_2)\, J_0(\lambda r_i)\, d\lambda\,. \tag{1}$$

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Use one space after periods and colons. Hyphenate complex modifiers: "zero-field-cooled magnetization." Avoid dangling participles, such as, "Using (1), the potential was calculated." [It is not clear who or what used (1).] Write instead, "The potential was calculated by using (1)," or "Using (1), we calculated the potential."

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Appendix

Appendixes, if needed, appear before the acknowledgment.

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